

Riya Vachhani

Surat, Gujarat | +91-9714471170 | riya.vachhani0001@gmail.com | [GitHub](#) | [Linkedin](#)

PROFILE

Passionate Python Developer with a growing foundation in Artificial Intelligence and Machine Learning, seeking an internship or entry-level opportunity to apply practical skills in real-world projects. Proficient in Scikit-learn, Pandas, and NumPy, with a strong understanding of mathematics and algorithms that supports effective ML model development. Eager to contribute to a collaborative team environment that values innovation, continuous learning, and building meaningful technology.

EDUCATION

A.D. Patel Institute of Technology , Anand	2023 – 2027
B.Tech in Artificial Intelligence and Data Science	CGPA: 8.30
Ashadeep IIT, Surat	2023
Higher Secondary (GSEB Board)	Percentage: 60.23%
Ashadeep School, Surat	2021
Secondary School (GSEB Board)	Percentage: 93.13%

PROJECT

HIREFLOW – HR Onboarding System (Smart Automation)

Full Stack AI Application - Python, Django, SQLite, HTML, CSS, Bootstrap, JavaScript

- Built an end-to-end HR automation platform for employee onboarding, resume screening, and document verification via a unified dashboard.
- Developed an AI-based resume parser for skill extraction and job-role matching, reducing manual screening effort.
- Designed a Django backend with role-based access control (RBAC) to manage HR, manager, and employee workflows securely.
- Created a responsive dashboard with real-time analytics, onboarding tracking, and task management for improved HR visibility.

PROJECT VERACITY – SELF HEALING LLM HALLUCINATION FIREWALL

Python, FastAPI, Groq API, ChromaDB, Next.js, TypeScript

- Built a real-time middleware system intercepting LLM outputs token-by-token, detecting and auto-correcting hallucinations before they reach end users.
- Engineered entropy-based hallucination detection using Shannon entropy over per-token Groq logprobs, with a 4-stage async pipeline (Interceptor → NLI Sentinel → Vector Vault → Rewriter) achieving under 200ms latency and 72% compute savings.
- Developed a Next.js UI with live correction highlights and source tooltips - first system to fix LLM hallucinations during streaming rather than post-hoc.
- Solved enterprise AI reliability by eliminating confident hallucinations while fully preserving streaming UX, delivering corrected answers instead of uncertainty flags.

TECHNICAL SKILLS

- **Programming Languages** : Python, Django, HTML, CSS, JS
- **Frameworks & Libraries** : Pandas, NumPy, SciKit-learn, MERN Stack (MongoDB, Express.js, React.js, Node.js), Bootstrap
- **Core Competencies** : AI, Data Science, Machine learning, Probability & Statistics, Linear Algebra

CERTIFICATIONS

- **AI4Dev (2026)**: 3rd Runner-up (From 600 teams), PSG Tech - Coimbatore
- **XEN-O-THON (2026)**: Grand Finalist (Top 15 teams), Guru Tegh Bahadur Institute of Technology - Delhi
- **InnovAtion HACKATHON (2026)**: Grand Finalist, Dhirubhai Ambani University - Gandhinagar
- **SMART INDIA HACKATHON (2024 & 2025)**: Participation